

nmos $\frac{1}{\mu_n C_{ox}} \frac{W}{L}$ $\frac{1}{\mu_n C_{ox}} \frac{L}{W}$ pmos $\frac{1}{\mu_p C_{ox}} \frac{W}{L}$ $\frac{1}{\mu_p C_{ox}} \frac{L}{W}$

max: $V_{dd} - V_{th}$ min: $|V_{th}|$

pass 0's pass 1's

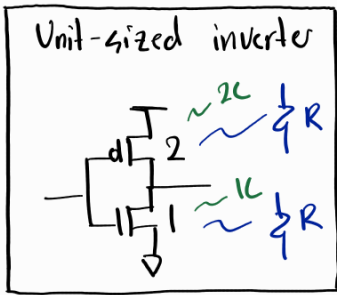
$\text{OR} = \bar{A} + \bar{B} = \overline{A \cdot B} = \text{NAND}$ $\text{NAND} = \overline{A \cdot B} = \bar{A} + \bar{B} = \text{OR}$

feature size f = channel length $L = 2\lambda$

"0.8 μm cmos" means $L = 0.8 \mu\text{m} = 2\lambda$

CMOS logic: PUN does Y (pmos)
PDN does $\bar{Y}$ (nmos)
Both networks get same inputs

	Cutoff	Triode	Saturation
C_{gb}	C_o	0	0
C_{gs}	0	$\frac{1}{2} C_o$	$\frac{2}{3} C_o$
C_{gd}	0	$\frac{1}{2} C_o$	0



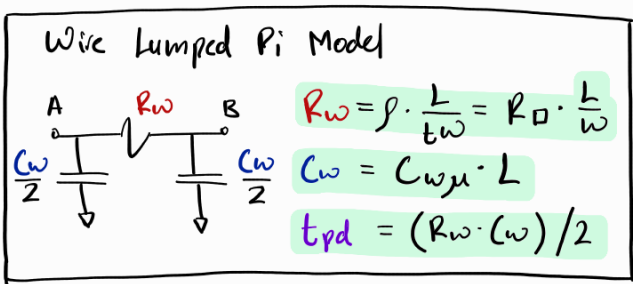
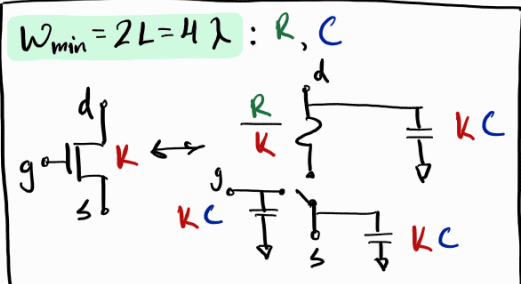
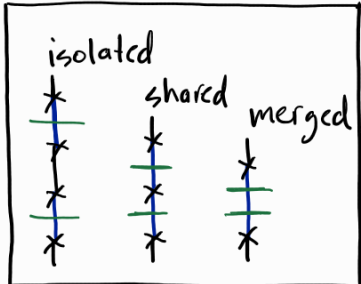
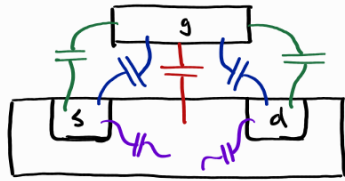
$t_{pd} = \sum \tau_k = \sum R_k \cdot C_k$ (shared path)

intrinsic gate capacitance: $C_o = C_{ox} \cdot W \cdot L = C_{gx} \cdot W$

gate overlap capacitance: $C_{gs/dot} = C_{gs/dot_{ox}} \cdot W$

junction/diffusion capacitance: $C_{s/d} = C_j(\text{area s/d}) + C_{jsw}(\text{perim. s/d})$

$C_{ox} = \frac{\epsilon_{ox} \cdot \epsilon_0}{t_{ox}}$



Basic Delay: $\tau_{inv} = R \cdot 3C_g$

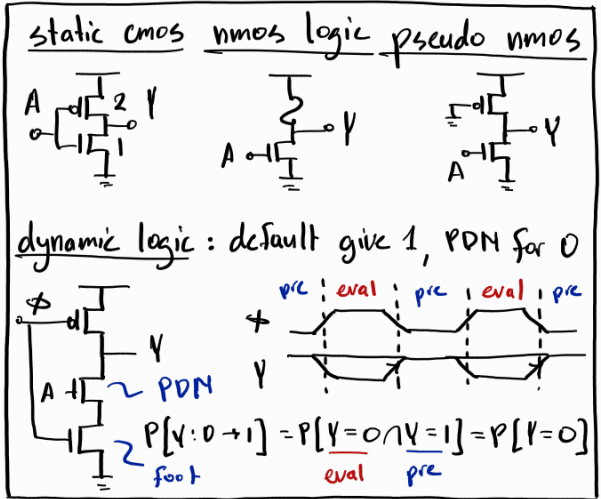
Normalized Delay: $d = \gamma + h = \frac{C_j}{C_g} + h = \frac{t_{pd}}{\tau_{inv}}$

Fanout: $h = \frac{C_{out}}{C_{in}}$

$D = d_1 + d_2 + d_3 + d_4 + d_5$

$= 4(\gamma + h) + (\gamma + \frac{C_L}{256C_{in}})$

Stages: $N = \frac{\ln(C_L/C_{in})}{\ln(h)}$, $h = \exp[\frac{\ln(C_L/C_{in})}{N}]$



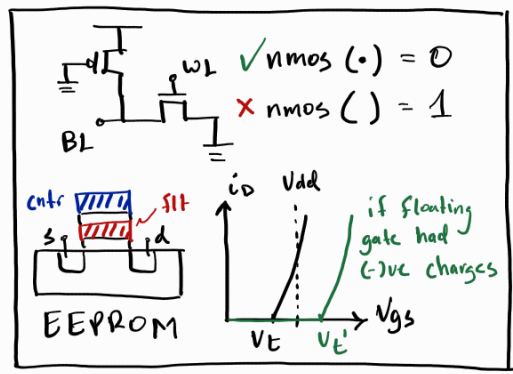
Max Delay Constraint: $T_c - t_{su} - t_{sk} \geq t_{pcq} + t_{pd}$

Min Delay Constraint: $t_{ccq} + t_{cd} \geq t_{hold} + t_{sk}$

Max to avoid su time viol. $\nearrow$ relaxes $T_c + t_{pdw}$

Min to avoid race cond. $\nwarrow$ tightens $T_c - t_{pdw}$

FF's $\rightarrow$ clk t_{pd} -wire: $\nearrow$ max constraint



generally: $\frac{V_{dd}}{2} + \frac{C_{cell}}{C_{BL} + C_{cell}} \cdot \frac{V_{dd}}{2} \Delta V$

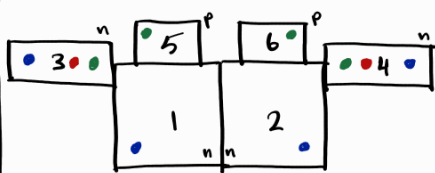
degraded case here: $\frac{1.8}{2} \times 450f + 1.4 \text{ or } 0 \times 50f$

450f + 50f

N transmissions/sec

Failure rate = $N \times P_{failure}$ [failures/sec]

MTBF = $\frac{1}{\text{failure rate}}$ [sec]



- 1) read stab: don't want to turn opp. nmos on ($V_a \leq V_t$), M_H sat, M_2 tri
- 2) read speed: wide enough to get ΔV in Δt ($\frac{I_{OH}}{C_{BL}} = \frac{\Delta V}{\Delta t}$), M_H sat, M_2 tri
- 3) write abil: want to turn opp. nmos off ($V_a \leq V_t$), M_6 sat, M_4 tri

dec. row, col.
from $n-k, k$
 $(n-k) = (k+m) = \text{square}$
 $\underbrace{2}_{\text{rows}} \times \underbrace{2}_{\text{words per row}} \times \underbrace{2}_{\text{bits per word}} = 2^{km}$

Triode: $i_D = \mu_{n/p} C_{ox} (w/L) [(V_{gs/sg} - |V_{tn/p}|) V_{ds/sd} - (V_{ds/sd})^2 / 2]$
Saturation: $i_D = (1/2) \mu_{n/p} C_{ox} (w/L) (V_{gs/sg} - |V_{tn/p}|)^2 [1 + \lambda V_{ds/sd}]$

$$R = \frac{2}{\mu_{n/p} C_{ox} (w/L) (V_{gs/sg} - |V_{tn/p}|)}$$

$P_{\text{switching}} = \alpha \cdot f \cdot C \cdot V_{dd}^2$
 $P[Y:0 \rightarrow 1]; \lambda_{\text{nand}} = \lambda_{\text{nor}} = 3/16$

50% $\leftrightarrow 0.69 \tau$, 63% $\leftrightarrow 1 \tau$, 90% $\leftrightarrow -\ln(1/10) \tau = 2.303 \tau$

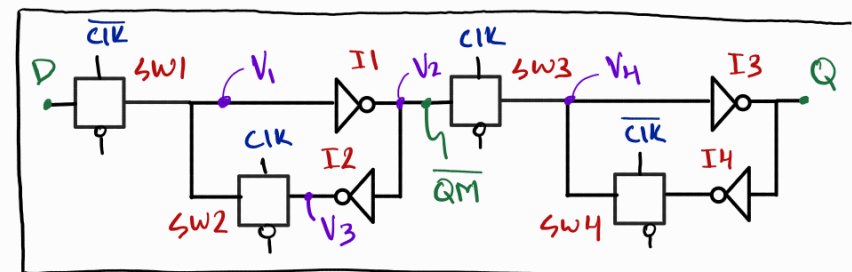
Combinational: t_{pd}, t_{cd}
Latches: $t_{\text{setup}}, t_{\text{hold}}, t_{\text{pcq}}, t_{\text{ccq}}, t_{\text{pdq}}, t_{\text{cdq}}$
FF's: $t_{\text{setup}}, t_{\text{hold}}, t_{\text{pcq}}, t_{\text{ccq}}$

propagation delay t_{pd} : worst case delay
contamination delay t_{cd} : best case delay
fanout h : $\neq$ identical gates as load

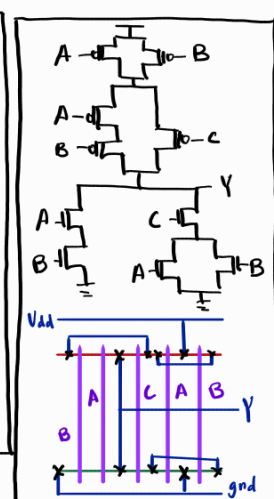
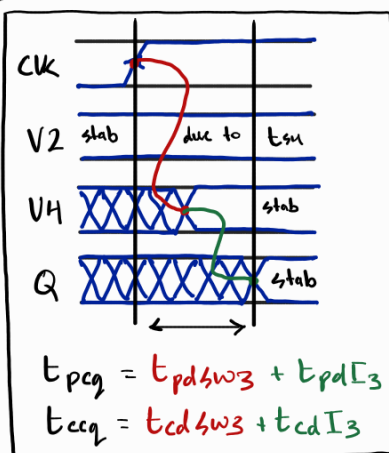
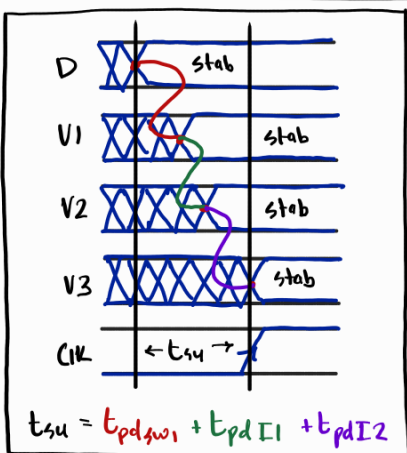
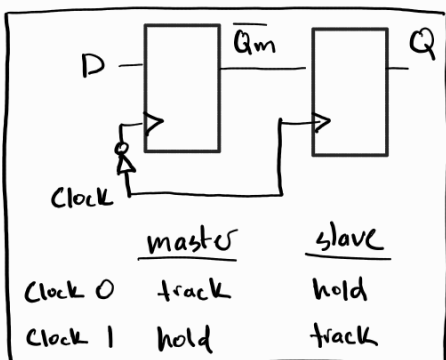
$P_{\text{failure}} = \frac{T_0 \sim \text{rise time}}{T_c \times \exp(T_c / \tau_s)}$
(clock period - T_{su}) $\frac{1}{g_{mnp} t_{mp}}$

	Source	Drain	$V_{ov} = V_{gs} - V_t$
Cutoff (open switch)	$V_{gs} \leq V_t$		
Triode (V. entr. R)	$V_{gs} > V_t$	$V_{gd} > V_t, V_{ds} < V_{ov}$	$\underbrace{M_n C_{ox} (w/L)}_{k_n'}$
Saturation (V. entr. I_s)	$V_{gs} > V_t$	$V_{gd} \leq V_t, V_{ds} \geq V_{ov}$	k_n

Max delay constraint: min time available $\geq$ max time required
Min delay constraint: min time data is stable $\geq$ max time must be stable



DRAM refresh:
 $I_{\text{leak}} = C_{\text{cell}} \frac{\Delta V}{\Delta t}$
 ΔV your tolerance
 Δt your refresh time



- mux out of transmission gates takes less T 's but is nonrestoring
- and out of nands/nois takes more T 's but is restoring
- t_{pd} wire $\propto \text{length}^2$
- d/gram in V_{dd}
 - 1) folded
 - 2) open

